

Rest Web Service

The goal of this lab work is to develop a small application for cars renting.

The functionalities to be implemented are:

- Get a list of unrented cars
- Rent a car
- Get back a car

Programming language

Choose the language you want but Java is recommended (version JDK 17, 21 or 22).

Gradle project creation (if Java)

Use the Spring Initializer (<https://start.spring.io/>) to create a project:

- Don't forget to choose Gradle Project
- Give it a name (Artifact)
- Add Spring Web as dependencies (library)

Download and unzip the project. Then open the project inside Eclipse (import gradle project) or IntelliJ (open project).

Create a new class (along with the main program) into src/main/java/

```
import org.springframework.web.bind.annotation.*;
```

```
@RestController
public class HelloService {
    @GetMapping("/")
    public String hello() {
        return "hello";
    }
}
```

Launch the main program: `/src/main/java/package.../*Application.java`

Open your web browser at: <http://localhost:8080>

The car rental app

Code the following features (see examples at the next paragraph):

- Get the features of a car:

- URI: .../cars/plateNumber
- http GET
- Json response: { " plateNumber" : "11AA22" , " brand" : "Ferrari, "price" : 100 }
- Rent a car:
 - URI: .../plateNumber?rent=true
 - http PUT
 - Send Json inside the http body: { "begin" : "11/11/2017" , " end" : "1/1/2018" }
- Get back the car:
 - URI: .../plateNumber?rent=false
 - http PUT

Advice: you can use the following templates.

Http request: <http://localhost:8080/cars>

```
@GetMapping("/cars")
@ResponseStatus(HttpStatus.OK)
@ResponseBody
public List<Car> listOfCars(){
}
```

Http request: <http://localhost:8080/cars/AA11BB>

```
@GetMapping("/cars/{plateNumber}")
@ResponseStatus(HttpStatus.OK)
@ResponseBody
public Car aCar(@PathVariable("plateNumber") String plateNumber) throws Exception{
}
```

Http request: <http://localhost:8080/cars/AA11BB?rent=true>

```
@PutMapping(value = "/voiture/{plateNumber}")
@ResponseStatus(HttpStatus.OK)
public void rentOrGetBack(
    @PathVariable("plateNumber") String plateNumber,
    @RequestParam(value="rent", required = true)boolean rent) throws Exception{
}
```

Http request: <http://localhost:8080/cars/AA11BB?rent=true>

With a class as a parameter sent in the request body:

```
@PutMapping(value = "/cars/{plateNumber}")
public void rent(
    @PathVariable("plateNumber") String plateNumber,
    @RequestParam(value="rent", required = true)boolean rent,
    @RequestBody Dates dates){
}
```